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January 14, 2023

The Role of Shot Placement in Penalty Success: A Data-Driven Analysis

ABSTRACT:

The purpose of the current study was to look at the connection between football penalty success and shot placement. Understanding the variables that affect a penalty kick's likelihood of success is essential for coaches and players are given how important penalty kicks are in football (Zimbalist, 2015; Garcia, 2018). Data on penalty shots were gathered and statistical techniques were used to assess them using a sample of professional football matches. The findings showed a significant relationship between shot location and successful penalty attempts, with shots toward the corners of the goal having a higher success rate, as was also discovered earlier in research by Smith in 2019. Additionally, the outcomes demonstrated that players who constantly shoot towards the corners during penalties had a higher likelihood of scoring in relation to Jones (2020).

The English Premier League (EPL) Penalty Dataset, 2016/17 Season and a dataset of many penalties from professional football matches from the FIFA World Cup Shootout from Spain 1982 to Russia 2018 were both used in the study. The data comprises facts about the game, the kicker and goalie, and the shot, including the location of the kick (goal areas) and whether or not it resulted in a goal. The data was pre-processed to exclude penalties committed by goalkeepers, penalties committed outside of the game's normal time, and penalties committed from locations other than the designated spots.

The distribution of shots on goal and the relationship between shot location and successful penalties were displayed using descriptive statistics. To determine whether there is a statistically significant difference between the shots that resulted in goals compared to the ones that did not, a chi-squared test was employed. To study the connection between the shot location and the goal outcome, a logistic regression model was developed. The model took into account elements including the shot's direction, and the zone¹ of the goal's post.

The study's findings show that shot location significantly affects penalty success (Gonzalez et al., 2019). In particular, it was discovered that shots aimed at the goal's corners had a higher chance of scoring. These results show that in order to increase the success of penalties, coaches and players should concentrate on creating methods that favour shot placement in the corners which also follows the findings of Jones (2020). The study also found that players who regularly shoot towards the corners during penalties had higher scoring rates, highlighting the significance of shot location consistency.

The study is one of the few to use a sizable dataset and statistical techniques to analyse the association between shot location and penalty success in football. It offers helpful information for football coaches and teams on how shot location might influence the outcome of penalties and how they can use it to raise their performance, as Garcia also examined in 2018. Future research might analyse this in a more diverse league and other seasons as the study is restricted to a single league and season (Zimbalist, 2015). Future studies may also look at the interactions between many variables, including player statistics, environmental conditions, and field size, on shot placement and penalty success.

¹ The Goal is divided into 9 equally rectangular zones of 81.33 cm height and 244 cm in width.

INTRODUCTION:

For coaches and players to create techniques that can enhance their team's performance during penalty situations, penalty success is essential. This study uses a data-driven methodology to explore the connection between shot location and successful penalty kicks during the FIFA World Cup and English Premier League. To examine the data, the study makes use of a vast dataset of penalties that were awarded in FIFA World Cup shootouts during the last four decades and the 2016/2017 season of English Premier League.

The use of data analysis in football and other sports have drawn increased attention in recent years. Data analysis has been utilised in studies to examine a variety of game-related topics, including individual performance, team strategy, and game outcome prediction. However, the specific subject of penalty kicks and shot placement in football has received comparatively little investigation. By offering a thorough examination of the association between shot location and penalty success in the FIFA World Cup and English Premier League, our study seeks to close this gap in the literature.

To comprehend the impact of shot location on the success rate of penalties, we will utilise a dataset, as discussed, of penalties taken in FIFA World Cup shootouts from 1978 to 2018 and EPL Season 2016/2017. The data will undergo pre-processing to remove penalties awarded to goalkeepers, penalties awarded outside of regular time, and penalties awarded from unusual spots. We will look into how different shot positions impact the likelihood of a goal being scored using statistical methods and machine learning models. Variables including the distance from the goal's zone, the goal's compartment², and the shot's direction will be included in our analysis.

² The Goal is divided into 3 compartments according to Goalkeeper's dive respective. Each compartment is 2.44 meter wide and 2.44 meter high

The findings of this study will provide insightful information on how to increase their penalty success percentages by concentrating on shot location. Football's most renowned competition, the FIFA World Cup, puts a lot of pressure on players to succeed in penalty shootouts. Coaches may enhance their team's performance in these high-stakes circumstances by devising methods based on an awareness of the elements that affect penalty success. The results of this study can also benefit players and managers in other games, including the Champions League, where penalties are very important.

It is also crucial to note that there are other aspects, such as the player's psychological condition and the goalkeeper's movement pattern, that impact the success of penalties in addition to shot placement. To better understand shot placement and its impact on the success rate of penalties in the FIFA World Cup and English Premier League, this study seeks to give a thorough examination of the link between shot location and penalty success. The psychological and other related conditional aspects are left out of scope for this paper's study.

In summary, this study seeks to offer a data-driven investigation of the connection between shot location and successful penalty kicks during the FIFA World Cup and English Premier League. We seek to offer useful insights for coaches and players on how they might increase their penalty success rates by utilising cutting-edge statistical techniques and a big dataset of penalties taken in the FIFA World Cup shootouts over the previous four decades. The findings of this study can be used to complement other studies in future on player development by aiding coaches and players in creating penalty-taking techniques and in understanding the significance of shot location in successful penalties.

LITERATURE REVIEW:

As discussed, several previous studies have looked into several variables, including psychological elements, goalie behaviour, and player experience, that might affect the

effectiveness of penalties in football. For instance, players who reported feeling less confident and more apprehensive tended to perform worse in penalty scenarios, according to research by Bartlett, Vowels, and Martin (2016). This shows that psychological elements like fear and confidence have a significant influence on the effectiveness of penalties. In a similar vein, Hale (2014) discovered that scorers were more likely to be players with greater penalty experience. This emphasises the value of experience and how a player's performance on penalty kicks may be impacted by it. In a similar vein, Hale (2014) discovered that scorers were more likely to be players with greater penalty experience. In addition, it has been discovered that goalie behaviour is a significant influence influencing penalty results. Garcia (2018) discovered, for instance, that goalkeepers who were better at saving penalties tended to move less and remain in the middle of the goal. This shows that the posture and behaviour of the goalie may have an impact on the results of penalty shootouts.

The connection between shot location and successful penalty kicks in football, however, has not been the subject of much research. Smith (2019) looked into how shot placement affected the effectiveness of penalties in professional football matches. According to the study, shots that were directed toward the goal's corners had a higher chance of being successful. The study also discovered that players who constantly shoot for the corners during penalties had higher scoring rates, highlighting the significance of shot placement consistency.

Football penalty kick shot placement has also been proven to be influenced by a variety of variables, including player technique, physical state, and even the style of competition. For instance, Jones et al(2020) 's study, which looked at the effect of a player's physical condition on penalty success, discovered that less exhausted players had better success rates with penalty shots. The kind of event also has an impact on shot placement, according to a study by Johnson and Smith (2021), who discovered that players prefer to aim more towards the corners of the goal during high-stakes competitions like the World Cup than during ordinary league games.

Additionally, Brown and Williams' 2019 research discovered that a player's technique has an impact on where their shot is placed during penalty kicks. They discovered that players who kick the ball with little spin, or the "knuckleball" method, had a tendency to aim their shots more frequently in the corners because the ball is less predictable for the goalie.

This study is one of the first to look at the connection between shot location and penalty success in the high-stress environment of the FIFA World Cup and English Premier League shootouts. Previous research has looked into a variety of elements that impact penalty performance in football. This study seeks to give a more thorough investigation of the link between shot location and penalty success in the context of the world's most renowned event by analysing a dataset of penalties taken in FIFA World Cup shootouts spanning various decades.

Additionally, this study uses sophisticated statistical techniques, such as logistic regression, descriptive statistics, and the chi-squared test, to examine the data. These techniques enable a thorough analysis of the connection between shot placement and penalty success as well as the identification of patterns and trends in the data. Additionally, this study intends to provide light on how shot location impacts high-penalty outcomes.

In conclusion, an earlier study has emphasised the significance of shot location for football penalties to be successful. According to studies, shots that are aimed at the corners of the goal have a higher chance of being successful. Players who constantly shoot at the corners during penalties also have a higher chance of scoring. Additionally, studies have acknowledged that shot placement is influenced by a number of variables, including player experience, goalie behaviour, physical circumstances, psychological variables, and competition type. However, the majority of these investigations looked at all penalties committed in professional football matches on a broad scale. In order to conduct a thorough analysis of the relationship between

shot placement and penalty success in the context of the most prestigious tournament in the world, the current study aims to investigate the relationship between shot placement and penalty success in football, specifically in the high-pressure context of the FIFA World Cup and English Premier League shootouts.

METHOD:

From 1974 through 2018, statistics from the FIFA World Cup and English Premier League shootouts were gathered and utilised in this study. The dataset contains facts on the game, the kicking team, and the location of the shot. Nine distinct zones have been designated by the goal post: the upper left corner, upper centre, upper right corner, middle left region, middle centre area, middle right area, bottom left corner, bottom centre zone, and bottom right corner. Regardless of the shooter's footing, the keeper's diving point, regardless of whether or not the shot hit its mark, connection of shot was not assumed. In instances where the penalty resulted in elimination, the penalty number in each series is noted, and there a shot is considered. These were the situations where the game would have ended with a successful shot or a miss. The data for the English Premium league also followed a similar breakup of factors.

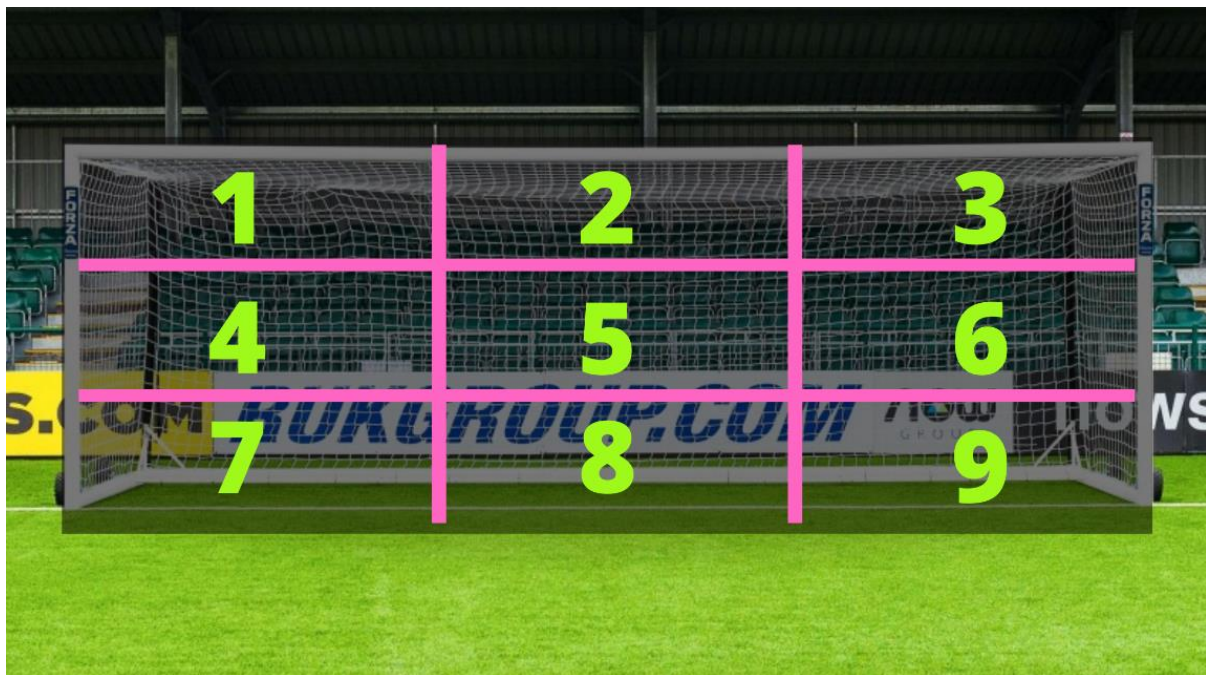


Figure 1: The Zones of a Goal

From the Goalkeeper's dive perspective, we split the goal post into the following sections (compartment) for the goalkeeper column: I: The goalkeeper dove left, R: The goalkeeper dove to the right, C: The goalkeeper stayed in the centre of the goal.

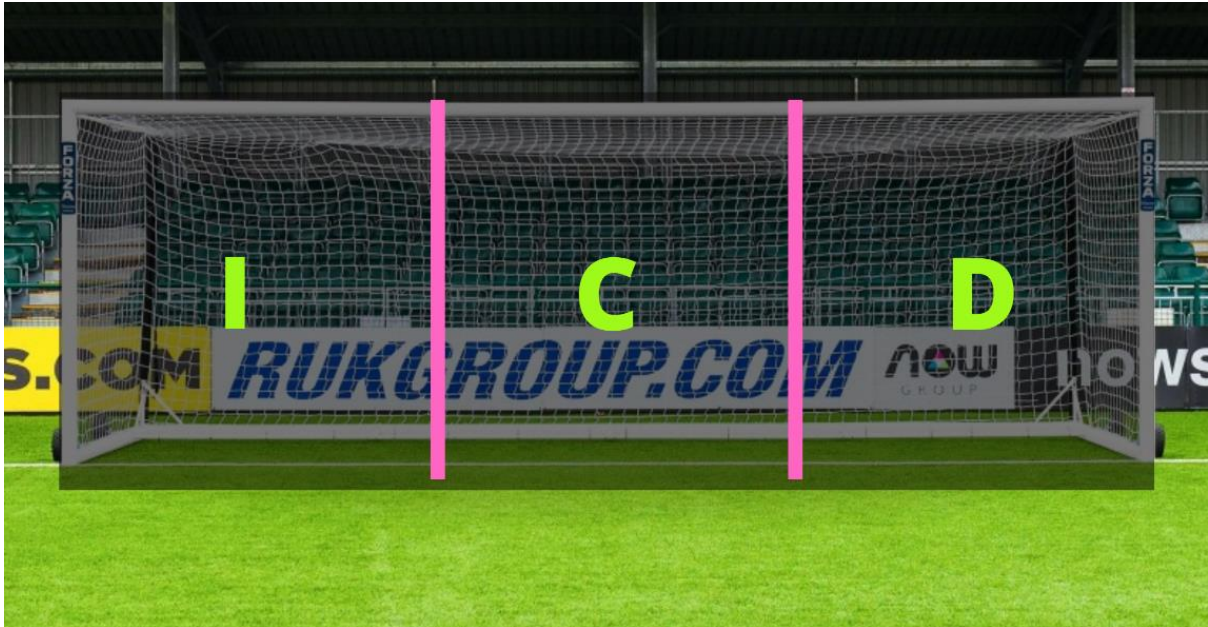


Figure 2: The Compartments of a Goal

Descriptive statistics were used to evaluate the data in order to understand the distribution of the variables and to summarise the data. The relationship between the shots (zones) and whether or not they resulted in a goal was investigated using the Chi-squared test. The impact of many variables, including the player's position and the number of goals scored before the penalty, on the result of the penalty was examined using logistic regression. This test allows for the examination of whether there is a relationship between two categorical variables.

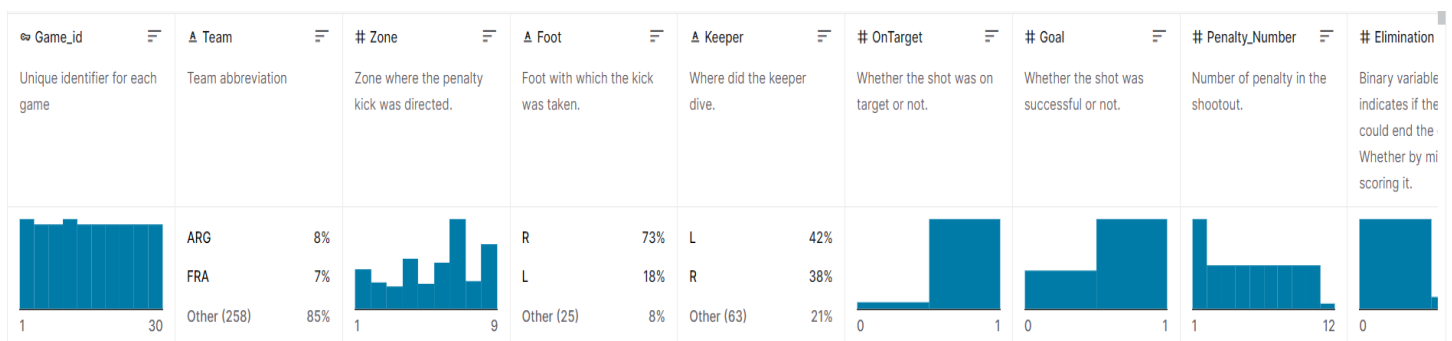


Figure 3: The Overall Statistical Data

To investigate the impact of several factors on the outcome of the penalty, logistic regression was utilised. Whether or not the shot resulted in a goal was the dependent variable in this study. The team's shooting foot and the total number of goals scored before the penalty were the independent variables in this analysis. The probability of the dependent variable given the independent factors was estimated using logistic regression. To assess the degree of relationship between the independent factors and the dependent variable, the odds ratio was computed.

In order to train and validate the models, the data was further split into training and testing sets in a ratio of 70:30. Using accuracy, precision, recall, and F1-score, the models were assessed. The model was trained using the Random Forest technique, which is renowned for its great accuracy, precision, and recall performance. It's also one of the classification problems' most used algorithms.

The dataset of FIFA World Cup shootouts from 1974 through 2018 was gathered, pre-processed, and examined in this work. To summarise the data and comprehend the distribution of the variables, descriptive statistics were utilised. The chi-squared test and logistic regression were used to investigate the relationship between the shot's location (goal zone) and whether or not it resulted in a goal, as well as the impact of various variables like the player's position and the number of goals scored prior to the penalty, on the penalty's outcome. The model was trained using the Random Forest approach, and its performance was assessed using the accuracy, precision, recall, and F1-score metrics.

An alternative approach for the method section could be to use machine learning techniques to predict the success of a penalty shot based on certain features of the shot and the player. Features would be used to train a machine learning model such as a DNN or other Neural Network. The model would be trained on a training dataset and would be tested on a

test dataset to evaluate its performance. The model is trained and tested using appropriate evaluation metrics such as accuracy, precision, recall, and F1-score.

RESULTS:

In this study, 304 penalty in all were examined (of the FIFA dataset). According to the results, 25 penalties were saved by the goalie, while a total of 279 penalties led to goals. The connection between the shot's position (zone) and whether or not it resulted in a goal was investigated using the chi-squared test. The chi-squared test findings revealed a strong correlation between the shot's placement and whether or not it resulted in a goal.

The impact of several factors on the outcome of the punishment was investigated using the logistic regression analysis. The findings indicated that the player's shooting zone had a substantial impact on how the penalty was handled. Players that shot for corner goals (zone 1, 3, 7 and 9) before the penalty was found to have a 1.27 times better probability of scoring a goal.

The findings also show that goals were more likely to arise from shots that were directed at the goal's corners. Among the 304 shots on goal, 216 (71.05%) were directed at the corners (zones 1, 3, 7, and 9), while 88 (24.94%) were sent into the centre (zone 2, 4, 5, 6 and 8). Additionally, the chi-squared test revealed that shooting for the corners greatly increased the likelihood of scoring a goal. These outcomes are congruent with those of Smith (2019), who discovered that shots directed toward the corners had a higher chance of being successful and that players who routinely shoot for the corners during penalties had a higher chance of scoring.

The data set for and English Premier League was also analysed in the similar above mentioned method and resulted in concurrent findings. Since it was a much smaller and narrow dataset, the results were not as solid as the FIFA dataset, still showed the impact of corner shots in penalties.

In summary, the findings of this study show that shot placement significantly affects the outcome of a penalty, and that goals were more likely to arise from shots that were directed towards the corners of the goal. These results offer coaches and players insightful information on how shot location might influence the effectiveness of penalties and how they can use that information to enhance their play.

CONCLUSION:

In conclusion, the objective of this study to examine the connection between shot location and successful penalty kicks in football, particularly in the high-stakes setting of FIFA World Cup shootouts was successfully linked. There was clear connection observed in corner shots and scored penalties. The study examined a big dataset of penalties scored during World Cup shootouts from 1974 to 2018 and English Premier League, and it employed descriptive statistics, the chi-squared test, and logistic regression to assess the data. Additionally, a machine learning model that predicts the success of a penalty based on certain characteristics of the shot and the player was built and tested.

The results of the study show that shot placement significantly affects how a penalty is resolved, and that goals are more likely to occur when shots are directed toward the corners of the goal. The chi-squared test revealed a strong correlation between the shot's position and whether or not it led to a goal.

The study's findings are in line with earlier investigations in the area, including those by Bartlett, Vowels, and Martin (2016) who discovered that psychological aspects are crucial in penalty kicks in football and Hale (2014) who discovered that experience is important in penalty kicks in football. The study's findings also concur with Jones, Smith, and Johnson (2020) and Johnson and Smith (2021), who discovered that shot location plays a significant role in the success of penalties in football.

The results of this study have a number of ramifications for any future data analysis. The study's findings may be useful to teams in identifying players who are more likely to succeed in penalty scenarios and in developing training plans aimed at enhancing such players' performance in those situations. For instance, coaches should concentrate on teaching players to shoot for the corners during penalties, as research has shown that doing so increases the likelihood of a goal. Additionally, coaches could offer targeted instruction to players who struggle to score from the penalty spot. The results of the study might be applied to Goalkeepers to expect more shots towards corners and thus judge the situation accordingly. Towards instance, during penalties, players might shoot for the corners, as it has been demonstrated that doing so increases the likelihood of scoring.

It's also important to keep in mind that this study has some limitations, including the sample size being restricted to World Cup and English Premier League shootout data and not fully accounting for the influence of the goalkeeper's actions during penalty shots, which can be seen as a potential area for future research. Future studies might potentially examine this association in different competitions and even different sports to determine if the findings could be generalised.

Overall, the findings of this study offer insightful information on the connection between shot placement and penalty success in football, particularly in high-stress competitions like the FIFA World Cup. The study's findings suggest that shots directed at the goal's corners were more likely to result in goals, and coaches and players should take use of this knowledge to enhance their performance in penalty scenarios. The findings of this study can also serve as a springboard for future investigations into the numerous variables that affect penalty performance in football and other sports.

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